AMBA® AXI™ and ACE™ Protocol Specification

AXI3™, AXI4™, and AXI4-Lite™
ACE and ACE-Lite™

ARM®

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AMBA AXI and ACE Protocol Specification
AXI3, AXI4, and AXI4-Lite
ACE and ACE-Lite

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Release Information

The following changes have been made to this specification.

Change history

Date	Issue	Confidentiality	Change
16 June 2003	A	Non-Confidential	First release
19 March 2004	B	Non-Confidential	First release of AXI specification v1.0
03 March 2010	C	Non-Confidential	First release of AXI specification v2.0
03 June 2011	D-2c	Non-Confidential	Public beta draft of AMBA AXI and ACE Protocol Specification
28 October 2011	D	Non-Confidential	First release of AMBA AXI and ACE Protocol Specification

Issues B and C of this document included an AXI specification version, v1.0 and v2.0. These version number have been discontinued, to remove confusion with the AXI versions, AXI3 and AXI4.

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Contents
AMBA AXI and ACE Protocol Specification AXI3, AXI4, and AXI4-Lite ACE and ACE-Lite

Preface
About this specification x
Using this specification xi
Conventions xiii
Additional reading xv
Feedback xvi

Part A AMBA AXI3 and AXI4 Protocol Specification

Chapter A1 Introduction
A1.1 About the AXI protocol A1-20
A1.2 AXI revisions A1-21
A1.3 AXI Architecture A1-22
A1.4 Terminology A1-25

Chapter A2 Signal Descriptions
A2.1 Global signals A2-28
A2.2 Write address channel signals A2-29
A2.3 Write data channel signals A2-30
A2.4 Write response channel signals A2-31
A2.5 Read address channel signals A2-32
A2.6 Read data channel signals A2-33
A2.7 Low-power interface signals A2-34

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Chapter A3 **Single Interface Requirements**
A3.1 Clock and reset A3-36
A3.2 Basic read and write transactions A3-37
A3.3 Relationships between the channels A3-40
A3.4 Transaction structure A3-44

Chapter A4 **Transaction Attributes**
A4.1 Transaction types and attributes A4-58
A4.2 AXI3 memory attribute signaling A4-59
A4.3 AXI4 changes to memory attribute signaling A4-60
A4.4 Memory types A4-65
A4.5 Mismatched memory attributes A4-69
A4.6 Transaction buffering A4-70
A4.7 Access permissions A4-71
A4.8 Legacy considerations A4-72
A4.9 Usage examples A4-73

Chapter A5 **Multiple Transactions**
A5.1 AXI transaction identifiers A5-76
A5.2 Transaction ID A5-77
A5.3 Transaction ordering A5-78
A5.4 Removal of write interleaving support A5-81

Chapter A6 **AXI4 Ordering Model**
A6.1 Definition of the ordering model A6-84
A6.2 Master ordering A6-85
A6.3 Interconnect ordering A6-86
A6.4 Slave ordering A6-87
A6.5 Response before final destination A6-88

Chapter A7 **Atomic Accesses**
A7.1 Single-copy atomicity size A7-90
A7.2 Exclusive accesses A7-92
A7.3 Locked accesses A7-95
A7.4 Atomic access signaling A7-96

Chapter A8 **AXI4 Additional Signaling**
A8.1 QoS signaling A8-98
A8.2 Multiple region signaling A8-99
A8.3 User-defined signaling A8-100

Chapter A9 **Low-power Interface**
A9.1 About the low-power interface A9-102
A9.2 Low-power clock control A9-103

Chapter A10 **Default Signaling and Interoperability**
A10.1 Interoperability principles A10-110
A10.2 Major interface categories A10-111
A10.3 Default signal values A10-112

Part B **AMBA AXI4-Lite Interface Specification**

Chapter B1 **AMBA AXI4-Lite**
B1.1 Definition of AXI4-Lite B1-122
B1.2 Interoperability B1-124
B1.3 Defined conversion mechanism B1-125

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B1.4 Conversion, protection, and detection B1-127

Part C ACE Protocol Specification

Chapter C1 About ACE
C1.1 Coherency overview C1-132
C1.2 Protocol overview C1-134
C1.3 Channel overview C1-137
C1.4 Transaction overview C1-142
C1.5 Transaction processing C1-146
C1.6 Concepts required for the ACE specification C1-147
C1.7 Protocol errors C1-150

Chapter C2 Signal Descriptions
C2.1 Changes to existing AXI4 channels C2-152
C2.2 Additional channels defined by ACE C2-153
C2.3 Additional response signals and signaling requirements defined by ACE C2-155

Chapter C3 Channel Signaling
C3.1 Read and write address channel signaling C3-158
C3.2 Read data channel signaling C3-167
C3.3 Read acknowledge signaling C3-170
C3.4 Write response channel signaling C3-171
C3.5 Write Acknowledge signaling C3-172
C3.6 Snoop address channel signaling C3-173
C3.7 Snoop response channel signaling C3-176
C3.8 Snoop data channel signaling C3-180
C3.9 Snoop channel dependencies C3-182

Chapter C4 Coherency Transactions on the Read Address and Write Address Channels
C4.1 About an initiating master C4-184
C4.2 About snoop filtering C4-187
C4.3 State changes on different transactions C4-188
C4.4 State change descriptions C4-190
C4.5 Read transactions C4-191
C4.6 Clean transactions C4-197
C4.7 Make transactions C4-200
C4.8 Write transactions C4-202
C4.9 Evict transactions C4-206
C4.10 Handling overlapping write transactions C4-207

Chapter C5 Snoop Transactions
C5.1 Mapping coherency operations to snoop operations C5-210
C5.2 General requirements for snoop transactions C5-213
C5.3 Snoop transactions C5-219

Chapter C6 Interconnect Requirements
C6.1 About the interconnect requirements C6-226
C6.2 Sequencing transactions C6-227
C6.3 Issuing snoop transactions C6-229
C6.4 Transaction responses from the interconnect C6-232
C6.5 Interactions with main memory C6-234
C6.6 Other requirements C6-237
C6.7 Interoperability considerations C6-239

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Chapter C7 **Cache Maintenance**
C7.1 ARCACHE and ARDOMAIN requirements C7-242
C7.2 Other cache maintenance considerations C7-243

Chapter C8 **Barrier Transactions**
C8.1 About barrier transactions C8-248
C8.2 Barrier transaction signaling C8-249
C8.3 Barrier responses and domain boundaries C8-251
C8.4 Barrier requirements C8-254

Chapter C9 **Exclusive Accesses**
C9.1 About Exclusive accesses C9-260
C9.2 Role of the master C9-261
C9.3 Role of the interconnect C9-263
C9.4 Multiple Exclusive Threads C9-266
C9.5 Exclusive Accesses from AXI components C9-267
C9.6 Transaction requirements C9-268

Chapter C10 **Optional External Snoop Filtering**
C10.1 About external snoop filtering C10-270
C10.2 Master requirements to support snoop filters C10-272
C10.3 External snoop filter requirements C10-273

Chapter C11 **ACE-Lite**
C11.1 About ACE-Lite C11-276
C11.2 ACE-Lite signal requirements C11-277

Chapter C12 **Distributed Virtual Memory Transactions**
C12.1 About DVM transactions C12-280
C12.2 Synchronization message C12-281
C12.3 DVM transaction process and rules C12-282
C12.4 Physical and virtual address space size C12-284
C12.5 DVM transactions format C12-286
C12.6 DVM transaction restrictions C12-287
C12.7 DVM Operations C12-288

Chapter C13 **Interface Control**
C13.1 About the interface control signals C13-296

Part D Appendices

Appendix A **Revisions**

Glossary

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Preface

This preface introduces the *AMBA AXI and ACE Protocol Specification*. It contains the following sections:

- *About this specification* on page x
- *Using this specification* on page xi
- *Conventions* on page xiii
- *Additional reading* on page xv
- *Feedback* on page xvi.

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About this specification

This specification describes:

- the AMBA 3 AXI protocol release referred to as AXI3
- the AMBA 4 AXI protocol releases referred to as AXI4 and AXI4-Lite
- the AMBA 4 protocol releases referred to as ACE and ACE-Lite.

Intended audience

This specification is written for hardware and software engineers who want to become familiar with the *Advanced Microcontroller Bus Architecture* (AMBA) and design systems and modules that are compatible with the *Advanced eXtensible Interface* (AXI) protocol.

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Using this specification

The information in this specification is organized into parts, as described in this section.

Part A, AMBA AXI3 and AXI4 Protocol Specification

Part A describes the AXI3 and AXI4 releases of the *AMBA AXI Protocol Specification*. It contains the following chapters:

Chapter A1 *Introduction*

Read this for an introduction to the AXI architecture, and to the terminology used in this specification.

Chapter A2 *Signal Descriptions*

Read this for a description of the signals that are used by the AXI3 and AXI4 protocols.

Chapter A3 *Single Interface Requirements*

Read this for a description of the basic AXI protocol transaction requirements between a master and slave.

Chapter A4 *Transaction Attributes*

Read this for a description of the AXI protocol and signaling that supports system topology and system level caches.

Chapter A5 *Multiple Transactions*

Read this for a description of the AXI protocol and signaling that supports out-of-order transaction completion and the issuing of multiple outstanding addresses.

Chapter A6 *AXI4 Ordering Model*

Read this for a description of the AXI4 ordering model.

Chapter A7 *Atomic Accesses*

Read this for a description of the mechanisms that support atomic accesses.

Chapter A8 *AXI4 Additional Signaling*

Read this for a description of the additional signaling introduced in AXI4 to extend the application of the AXI interface.

Chapter A9 *Low-power Interface*

Read this for a description of the control interface that supports low-power operation.

Chapter A10 *Default Signaling and Interoperability*

Read this for a description of the interoperability of interfaces that use reduced AXI signal sets.

Part B, AMBA AXI4-Lite Interface Specification

Part B describes AMBA AXI4-Lite. It contains the following chapter:

Chapter B1 *AMBA AXI4-Lite*

Read this for a description of AXI4-Lite that provides a simpler control register-style interface for systems that do not require the full functionality of AXI4.

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Part C, ACE Protocol Specification

Part C describes the ACE protocol. It contains the following chapters:

Chapter C1 *About ACE*

This chapter gives an overview of system level coherency and the architecture of the *AXI Coherency Extensions* (ACE) protocol.

Chapter C2 *Signal Descriptions*

This chapter introduces the additional ACE interface signals.

Chapter C3 *Channel Signaling*

This chapter describes the basic channel signaling requirements on an ACE interface.

Chapter C4 *Coherency Transactions on the Read Address and Write Address Channels*

This chapter describes the transactions issued on the read address and write address channels.

Chapter C5 *Snoop Transactions*

This chapter describes the snoop transactions seen on the snoop address channel.

Chapter C6 *Interconnect Requirements*

This chapter describes the ACE interconnect requirements.

Chapter C7 *Cache Maintenance*

This chapter describes the ACE cache maintenance operations.

Chapter C8 *Barrier Transactions*

This chapter describes the ACE memory and synchronization barrier transactions.

Chapter C9 *Exclusive Accesses*

This chapter describes the ACE Exclusive Accesses to shareable memory.

Chapter C10 *Optional External Snoop Filtering*

This chapter describes using an external snoop filter in an ACE system.

Chapter C11 *ACE-Lite*

This chapter describes the ACE-Lite interface.

Chapter C12 *Distributed Virtual Memory Transactions*

This chapter describes *Distributed Virtual Memory* (DVM) transactions.

Chapter C13 *Interface Control*

This chapter describes the optional signals that can be used to configure the ACE interface.

Part D, Appendices

This specification contains the following appendices:

Appendix A *Revisions*

Read this for a description of the technical changes between released issues of this specification.

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Conventions

The following sections describe conventions that this specification can use:

- *Typographic conventions*
- *Timing diagrams*
- *Signals* on page xiv
- *Numbers* on page xiv

Typographic conventions

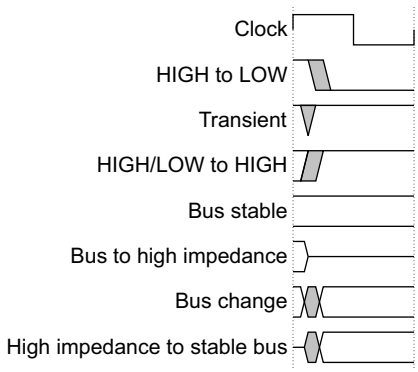
The typographical conventions are:

italic	Highlights important notes, introduces special terminology, and denotes internal cross-references and citations.
bold	Denotes signal names, and is used for terms in descriptive lists, where appropriate.
`monospace`	Used for assembler syntax descriptions, pseudocode, and source code examples. Also used in the main text for instruction mnemonics and for references to other items appearing in assembler syntax descriptions, pseudocode, and source code examples.
SMALL CAPITALS	Used for a few terms that have specific technical meanings.

Timing diagrams

The figure named *Key to timing diagram conventions* explains the components used in timing diagrams. Variations, when they occur, have clear labels. You must not assume any timing information that is not explicit in the diagrams.

Shaded bus and signal areas are undefined, so the bus or signal can assume any value within the shaded area at that time. The actual level is unimportant and does not affect normal operation.



Key to timing diagram conventions

Timing diagrams sometimes show single-bit signals as HIGH and LOW at the same time and they look similar to the bus change shown in *Key to timing diagram conventions*. If a timing diagram shows a single-bit signal in this way then its value does not affect the accompanying description.

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Signals

The signal conventions are:

Signal level The level of an asserted signal depends on whether the signal is active-HIGH or active-LOW. Asserted means:

- HIGH for active-HIGH signals
- LOW for active-LOW signals.

Lower-case n At the start or end of a signal name denotes an active-LOW signal.

Numbers

Numbers are normally written in decimal. Binary numbers are preceded by `0b`, and hexadecimal numbers by `0x`. Both are written in a `monospace` font.

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Additional reading

This section lists relevant publications from ARM.

See the Infocenter, http://infocenter.arm.com, for access to ARM documentation.

ARM publications

- *AMBA APB Protocol Specification* (ARM IHI 0024)
- *AMBA 4 AXI4-Stream Protocol Specification* (ARM IHI 0051)
- *ARM Architecture Reference Manual, ARMv7-A and ARMv7-R edition* (ARM DDI 0406)
- *ARMv7-M Architecture Reference Manual* (ARM DDI 0403)
- *ARMv6-M Architecture Reference Manual* (ARM DDI 0419).

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Feedback

ARM welcomes feedback on its documentation.

Feedback on this specification

If you have comments on the content of this specification, send e-mail to `errata@arm.com`. Give:

- the title, AMBA AXI and ACE Protocol Specification AXI3, AXI4, and AXI4-Lite ACE and ACE-Lite
- the number, ARM IHI 0022D
- the page numbers to which your comments apply
- a concise explanation of your comments.

ARM also welcomes general suggestions for additions and improvements.

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Part A

AMBA AXI3 and AXI4 Protocol Specification

Chapter A1
Introduction

This chapter introduces the architecture of the AXI protocol and the terminology used in this specification. It contains the following sections:

- *About the AXI protocol* on page A1-20
- *AXI revisions* on page A1-21
- *AXI Architecture* on page A1-22
- *Terminology* on page A1-25.

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